

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design and Develop a Responsive Webpage Using HTML5 and CSS3

2. AIM

To design and develop a simple, modern and responsive webpage using **HTML5 and CSS3**, containing a navigation bar, hero section, about section, services section, contact form and footer.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML5 webpage.
 - To create a webpage using HTML5 semantic elements.
 - To apply CSS3 styling to HTML elements.
 - To create a responsive navigation bar.
 - To design an attractive hero section using CSS.
 - To create service cards using Flexbox.
 - To style forms using CSS.
 - To implement hover effects and transitions.
 - To use CSS gradients for visual design.
 - To implement responsive design using media queries.
 - To make the webpage adaptable to different screen sizes.
 - To use <input>, <select> and <textarea> elements.
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4. THEORY

HTML5 is used to create the **structure and content** of a webpage, while CSS3 is used to control its **appearance, layout and presentation**.

In this experiment, HTML5 provides different sections of the webpage such as header, navigation, hero, about, services, contact and footer. CSS3 is then applied to style these sections and make the webpage visually attractive.

Important HTML Elements Used

1. <header>

The <header> element represents the top section of the webpage. In this webpage, it contains the website logo and navigation menu.

2. <nav>

The <nav> element contains links to different sections of the webpage such as Home, About, Services and Contact.

3. <section>

The <section> element is used to divide the webpage into different content sections.

4. <form>

The <form> element is used to collect user information such as name, email and message.

5. <input>

The <input> element is used to create text and email input fields.

6. <textarea>

The <textarea> element provides a multi-line input field for entering messages.

7. <button>

The <button> element creates the Send Message button.

8. <footer>

The <footer> element represents the bottom section of the webpage and contains copyright information.

CSS3 Concepts Used

1. Universal Selector

The * selector is used to apply common properties to all HTML elements.

2. Flexbox

Flexbox is used to arrange elements horizontally or vertically.

It is used in the header and service card layout.

3. Gradient

A linear gradient is used as the background of the hero section.

4. Hover Effect

The :hover pseudo-class changes the appearance of an element when the mouse pointer is placed over it.

5. Transition

The transition property creates a smooth animation when CSS properties change.

6. Box Shadow

The box-shadow property gives the service cards a shadow effect.

7. Media Query

Media queries are used to make the webpage responsive on smaller screens such as mobile phones and tablets.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file named index.html.
2. Create a CSS file named styles.css.
3. Add the basic HTML5 document structure.
4. Link the CSS file to the HTML file using the <link> element.
5. Create a header containing the website logo.
6. Add navigation links for Home, About, Services and Contact.
7. Create the Hero section with a heading, paragraph and Learn More button.
8. Create the About section and add information about the website.
9. Create the Services section.
10. Add three service cards for Web Design, Development and Support.
11. Use CSS Flexbox to arrange the service cards.

12. Create the Contact section.
 13. Add a form containing Name, Email, Message and Submit button.
 14. Create the Footer section with copyright information.
 15. Apply CSS styling to the webpage.
 16. Add colors, spacing, fonts, shadows and transitions.
 17. Add hover effects to navigation links, buttons and cards.
 18. Add a media query for responsive design.
 19. Open the HTML file in a web browser.
 20. Test the webpage on desktop and smaller screen sizes.
 21. Verify that all sections and navigation links work correctly.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">

  <title>My Webpage</title>

  <link rel="stylesheet" href="styles.css">

</head>

<body>

  <!-- Header -->
  <header>

    <div class="logo">
      MyWebsite
    </div>
```

```
<nav>

  <a href="#home">Home</a>

  <a href="#about">About</a>

  <a href="#services">Services</a>

  <a href="#contact">Contact</a>

</nav>

</header>

<!-- Hero Section -->
<section class="hero" id="home">

  <div class="hero-content">

    <h1>
      Welcome to My Website
    </h1>

    <p>
      A simple and modern webpage designed
      using HTML5 and CSS3.
    </p>

    <a href="#about" class="btn">
      Learn More
    </a>

  </div>

</section>
```

```
<!-- About Section -->
```

```
<section class="about" id="about">
```

```
  <h2>
```

```
    About Us
```

```
</h2>
```

```
  <p>
```

```
    We create clean, responsive and  
    user-friendly websites using modern  
    web technologies.
```

```
</p>
```

```
</section>
```

```
<!-- Services Section -->
```

```
<section class="services" id="services">
```

```
  <h2>
```

```
    Our Services
```

```
</h2>
```

```
  <div class="service-container">
```

```
    <div class="card">
```

```
      <h3>
```

```
        Web Design
```

```
</h3>
```

```
      <p>
```

```
        Attractive and responsive website  
        designs for all types of devices.
```

```
</p>
```

```
    </div>
```

```
<div class="card">
```

```
  <h3>
```

```
    Development
```

```
  </h3>
```

```
  <p>
```

```
    Modern websites developed using  
    HTML5, CSS3 and JavaScript.
```

```
  </p>
```

```
</div>
```

```
<div class="card">
```

```
  <h3>
```

```
    Support
```

```
  </h3>
```

```
  <p>
```

```
    Reliable maintenance and support  
    for your website.
```

```
  </p>
```

```
</div>
```

```
</div>
```

```
</section>
```

```
<!-- Contact Section -->
```

```
<section class="contact" id="contact">
```

```
  <h2>
```

```
    Contact Us
```

```
  </h2>
```

```
<form>

  <input type="text"
    placeholder="Your Name"
    required>

  <input type="email"
    placeholder="Your Email"
    required>

  <textarea
    placeholder="Your Message"
    rows="5">
  </textarea>

  <button type="submit">
    Send Message
  </button>

</form>

</section>

<!-- Footer -->
<footer>

  <p>
    &copy; 2026 MyWebsite.
    All Rights Reserved.
  </p>

</footer>

</body>

</html>
```

styles.css

```
*{
  margin: 0;
  padding: 0;
  box-sizing: border-box;
  scroll-behavior: smooth;
}

body {
  font-family: Arial, sans-serif;
  line-height: 1.6;
  color: #333;
  background-color: #f5f5f5;
}

/* Header */

header {
  background-color: #222;
  color: white;

  display: flex;
  justify-content: space-between;
  align-items: center;

  padding: 20px 8%;

  position: sticky;
  top: 0;

  z-index: 1000;
}

.logo {
  font-size: 26px;
  font-weight: bold;
}
```

```
nav {
  display: flex;
  gap: 25px;
}

nav a {
  color: white;
  text-decoration: none;
  font-size: 16px;
  transition: 0.3s;
}

nav a:hover {
  color: #00adb5;
}

/* Hero Section */

.hero {
  height: 90vh;

  display: flex;
  justify-content: center;
  align-items: center;

  text-align: center;

  background: linear-gradient(
    135deg,
    #00adb5,
    #393e46
  );

  color: white;
  padding: 20px;
}
```

```
.hero-content {
  max-width: 700px;
}

.hero h1 {
  font-size: 50px;
  margin-bottom: 20px;
}

.hero p {
  font-size: 20px;
  margin-bottom: 30px;
}

.btn {
  display: inline-block;

  background-color: white;
  color: #222;

  padding: 12px 25px;

  border-radius: 5px;

  text-decoration: none;
  font-weight: bold;

  transition: 0.3s;
}

.btn:hover {
  background-color: #222;
  color: white;
}

/* Common Section Styles */

section {
```

```
padding: 70px 8%;
text-align: center;
}

section h2 {
  font-size: 35px;
  margin-bottom: 20px;
}

/* About */

.about {
  background-color: white;
}

.about p {
  max-width: 700px;
  margin: auto;
  font-size: 18px;
}

/* Services */

.services {
  background-color: #f5f5f5;
}

.service-container {
  display: flex;
  justify-content: center;

  gap: 25px;

  margin-top: 35px;

  flex-wrap: wrap;
}
```

```
.card {
  background-color: white;

  width: 300px;

  padding: 30px;

  border-radius: 10px;

  box-shadow:
    0 5px 15px rgba(0, 0, 0, 0.1);

  transition: 0.3s;
}

.card:hover {
  transform: translateY(-8px);
}

.card h3 {
  color: #00adb5;

  margin-bottom: 15px;

  font-size: 23px;
}

/* Contact */

.contact {
  background-color: white;
}

form {
  max-width: 600px;

  margin: 30px auto 0;
```

```
display: flex;

flex-direction: column;

gap: 15px;
}

input,
textarea {
  width: 100%;

  padding: 14px;

  border: 1px solid #ccc;

  border-radius: 5px;

  font-size: 16px;

  font-family: inherit;
}

textarea {
  resize: vertical;
}

button {
  padding: 14px;

  border: none;

  border-radius: 5px;

  background-color: #00adb5;

  color: white;

  font-size: 17px;
```

```
    cursor: pointer;

    transition: 0.3s;
}

button:hover {
    background-color: #008b91;
}

/* Footer */

footer {
    background-color: #222;

    color: white;

    text-align: center;

    padding: 20px;
}

/* Responsive Design */

@media (max-width: 768px) {

    header {
        flex-direction: column;
        gap: 15px;
    }

    nav {
        gap: 15px;

        flex-wrap: wrap;

        justify-content: center;
    }
}
```

```
}

.hero h1 {
  font-size: 36px;
}

.hero p {
  font-size: 17px;
}

section {
  padding: 50px 5%;
}

.service-container {
  flex-direction: column;
  align-items: center;
}

.card {
  width: 100%;
  max-width: 350px;
}
}
```

7. CODE EXPLANATION

Code	Explanation
<!DOCTYPE html>	Declares that the document uses HTML5..
<html>	Root element of the HTML document

<head>	Contains metadata and page information.
<title>	Specifies the title of the webpage.
<body>	Contains the visible webpage content.
<header>	Creates the header section of the webpage.
<nav>	Contains navigation links.
<link rel="stylesheet">	Connects the HTML file with the CSS file.
.logo	Styles the website name.
.hero	Creates the main hero/banner section.
.hero-content	Controls the width of hero content.
.btn	Styles the Learn More button.
:hover	Applies styling when the mouse pointer is placed over an element.
<label>	Provides a label for form controls.
<input>	Creates input controls such as text, email and buttons.
<select>	Creates a dropdown list.
<option>	Defines options inside the dropdown list.
<textarea>	Creates a multi-line text input
<footer>	Creates the footer section.
transition	Creates smooth visual changes.

box-shadow	Adds a shadow around service cards.
display: flex	Creates a flexible layout.
linear-gradient()	Creates a gradient background.
position: sticky	Keeps the header visible while scrolling.

8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Click Home	Home section should be displayed	Home section displayed	Pass
TC-02	Open index.html	Homepage should load successfully	Homepage loaded	Pass
TC-03	Open webpage on mobile/small screen	Layout should adjust responsively	Responsive layout displayed	Pass
TC-04	Enter valid name and email	Form should accept input	Input accepted	Pass

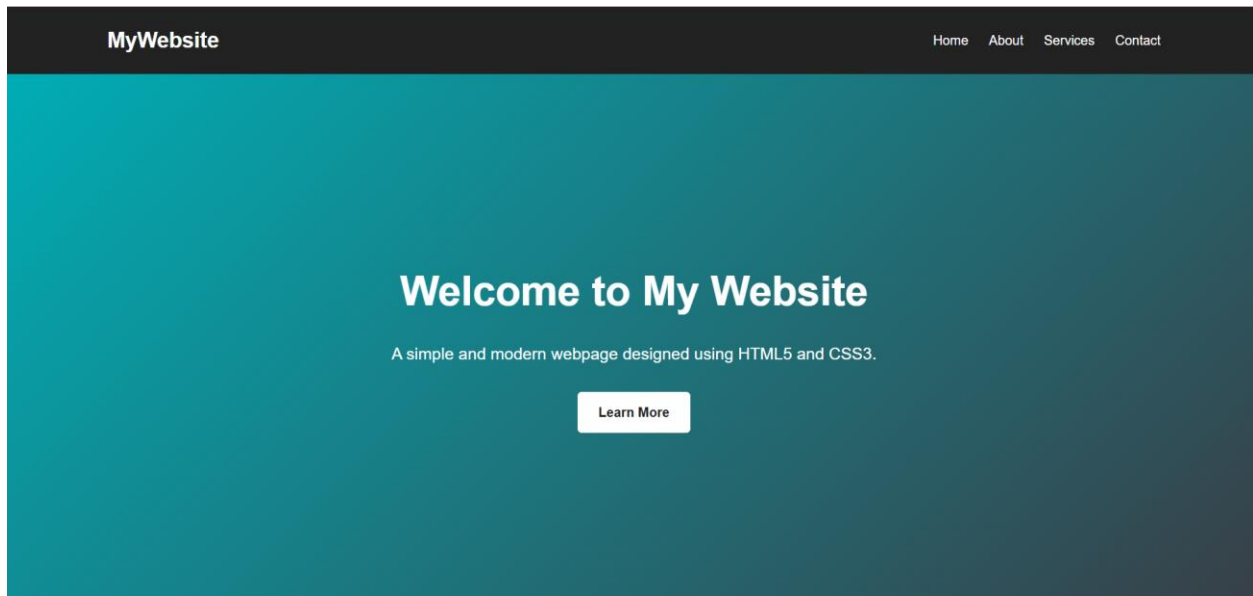
Additional Validation Test

Test Case	Input	Expected Result	Status
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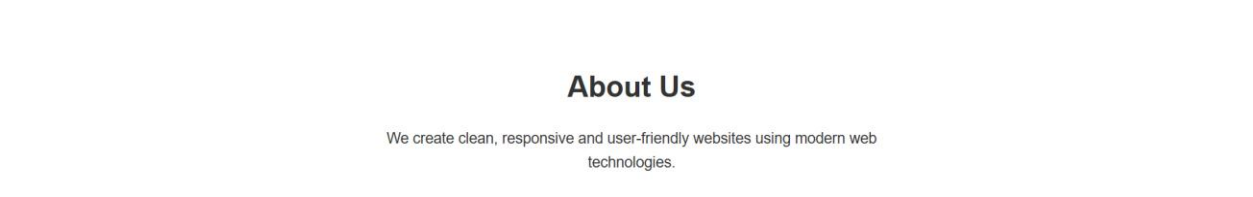
TC-05	Leave required fields empty	Browser should display validation	Pass
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9. OUTPUT

Output 1: Home Page



Output 2: About Section



Output 3: Services Section

Our Services

Web Design

Attractive and responsive website designs for all types of devices.

Development

Modern websites developed using HTML5, CSS3 and JavaScript.

Support

Reliable maintenance and support for your website.

Output 4: Contact Form

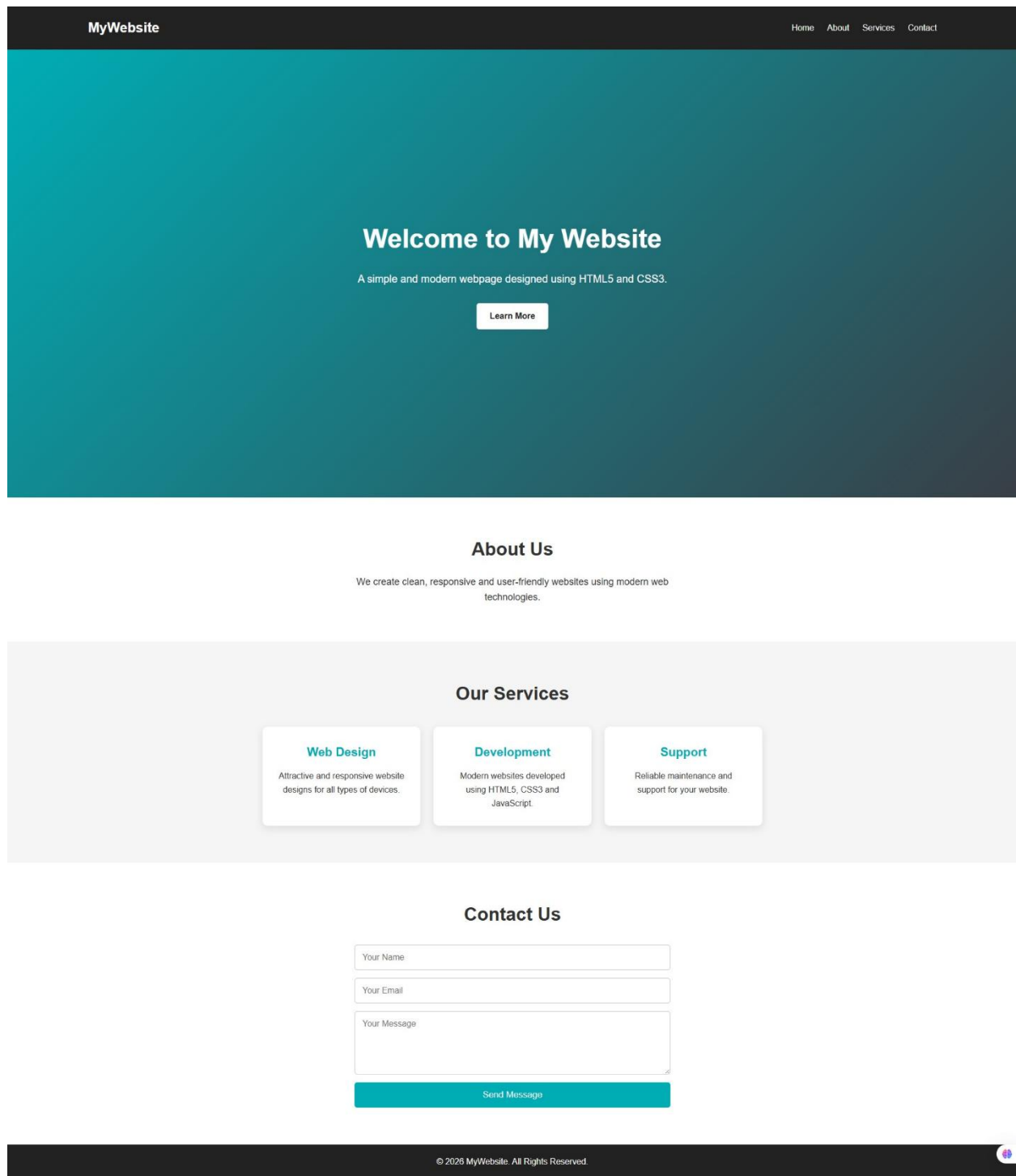
Contact Us

Your Message

Send Message

Output 5: Footer

Output 6: Whole website



10. RESULT

Thus, a **responsive and modern webpage using HTML5 and CSS3** was successfully designed and developed.

The webpage contains a navigation bar, hero section, about section, services section, contact form and footer. CSS3 features such as **Flexbox, gradients, transitions, hover effects, box shadows and media queries** were successfully implemented.

The webpage was tested on different screen sizes and was found to be responsive and user-friendly.
